

CIVICCONNECT
Civic & Public Grievance Management System
(Subject: Database Management System (DBMS))
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Class/Division: B.Tech Artificial Intelligence and Data Science, S.Y.
Academic Year: 2026–2027

Project Repository

GitHub — <https://github.com/avi-a11y/CivicConnect-DBMS-Project>

Project Technology

MySQL, Python, HTML5, CSS3, JavaScript

1. Introduction

CivicConnect is a database-centric DBMS mini project designed to manage student civic issues and public grievances in an organized manner.

The system stores information about students, departments, constituencies, MLAs, representatives, issue categories, issues, responses, assignments, notifications and status history using a relational MySQL database.

The project demonstrates important DBMS concepts such as tables, primary keys, foreign keys, CRUD operations, joins, subqueries, normalization, views, triggers, stored procedures and reporting.

2. Problem Statement

Students and citizens may face difficulty in reporting and tracking local civic issues such as infrastructure, sanitation, transportation and other public concerns.

Traditional methods can make it difficult to maintain complaint records, assign issues to representatives, track issue status and maintain response history.

CivicConnect provides a structured database system to store, manage and track civic issues efficiently.

3. Objectives

- Maintain student and department information.
- Manage constituency and representative information.
- Categorize civic and public issues.
- Allow users to submit and track issues.
- Assign issues to appropriate representatives.
- Maintain responses and status history.
- Manage notifications related to issues.
- Provide CRUD operations and useful reports.
- Reduce data redundancy through normalization.
- Demonstrate advanced DBMS features such as views, triggers and stored procedures.

4. Database Design

The project contains 11 related

Sr.No.	Table	Purpose
1	Departments	Stores departments information
2	Students	Stores student information
3	Constituencies	Stores constituencies details
4	MLAs	Stores MLA information
5	Representatives	Stores issue handling representatives
6	Issues Categories	Classifies different types of issues
7	Issues	Store submitted civic issues
8	Responses	Store responses to submitted issues
9	Status History	Tracks tables issues status changes
10	Notification	Store user notification
11	Assignments	Connects issues with representatives

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5. Key Relationships and Keys

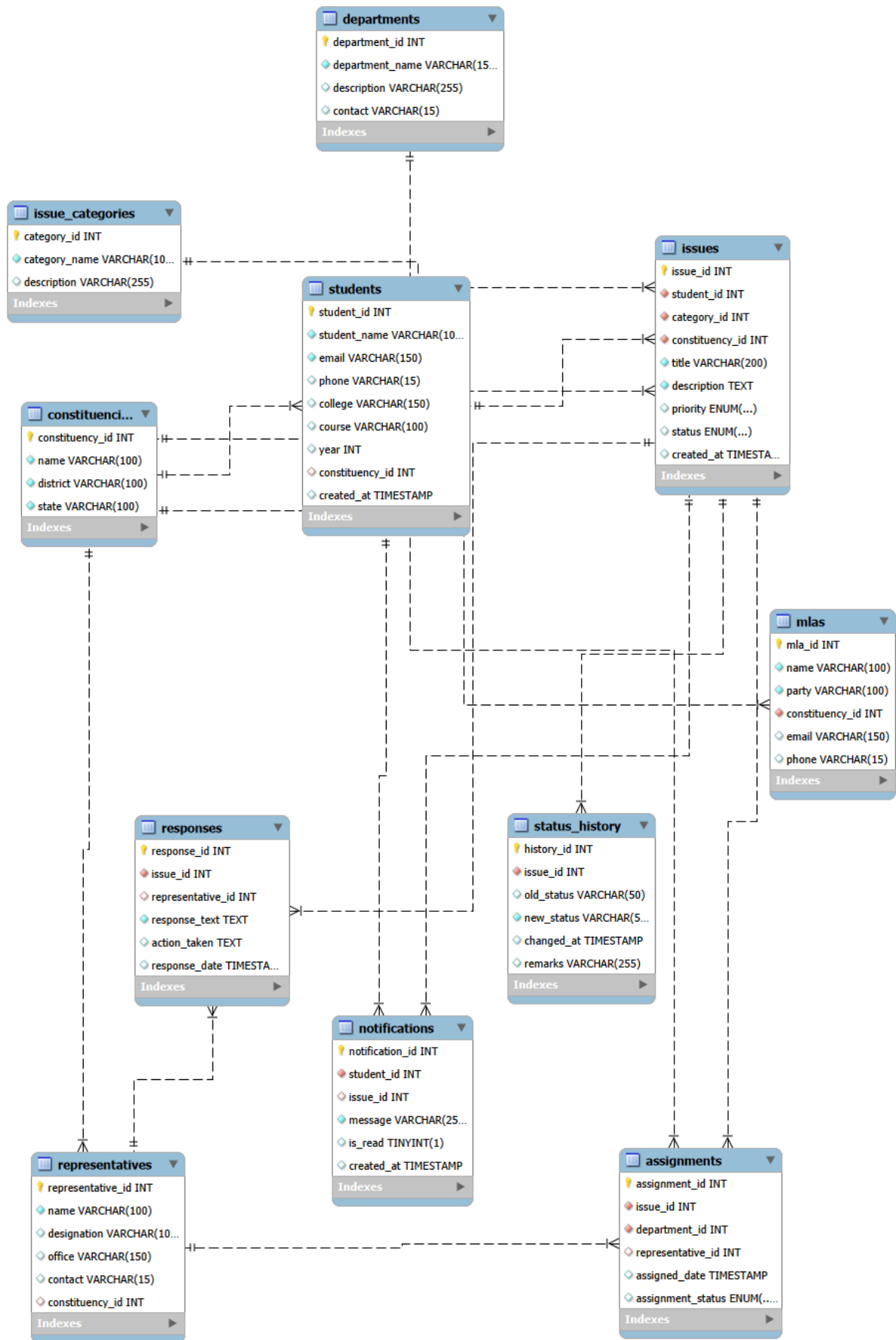
Primary keys uniquely identify records in each table, while foreign keys establish relationships between related tables.

Major Relationships

- Departments → Students: One department can have many students.
- Students → Issues: One student can submit multiple issues.
- Issue Categories → Issues: One category can contain multiple issues.
- Constituencies → Issues: One constituency can have multiple issues.
- Constituencies → MLAs: A constituency can be associated with an MLA.
- Issues → Responses: An issue can have multiple responses.
- Issues → Status History: An issue can have multiple status changes.
- Issues → Assignments: Issues can be assigned to representatives.
- Representatives → Assignments: A representative can handle multiple assigned issues.
- Students → Notifications: Notifications can be associated with students.

6. ER Diagram

The following ER diagram represents the major entities, attributes, primary/foreign keys and relationships described in the project



7. CRUD Operations

CRUD stands for Create, Read, Update and Delete.

Operation	Example
Create	Add a new student or submit a new civic issues
Read	Display issues , students and representatives
Update	Remove unwanted notification or record
Delete	Remove unwanted notification or records

8. Joins and Subqueries

Concept	Description	Example
JOIN	Combines data from multiple related tables	Displays issues with students, category and constituency
Subquery	A query written inside another SQL query	Find students with multiple issues

9. Normalization – 3NF

The database separates information into independent but related tables such as Students, Issues, Constituencies, Issue Categories, Representatives and Responses.

Normalization helps to:

- Reduce data redundancy.
- Avoid insertion, deletion and update anomalies.
- Improve data consistency.
- Maintain relationships using primary and foreign keys.

The database design follows the principles of Third Normal Form (3NF).

10. Advanced DBMS Features

10.1 View

A view can be created to display unresolved issues.

```
CREATE VIEW pending_issues AS
SELECT
    issue_id,
    title,
    priority,
    status,
    created_at
FROM issues
WHERE status <> 'Resolved';
```

This view provides a convenient way to access pending issues without repeatedly writing the complete query.

10.2 Trigger

A trigger can automatically record changes when an issue's status is updated.

Flow:

Issue Status Updated → Trigger Executes → Status History Updated

10.3 Stored Procedure

A stored procedure can be designed to retrieve issues for a particular constituency or student.

Flow:

Input ID → Find Related Issues → Display Issue Details

11. Search and Reports

The system can support searching by:

- Student Name
- Issue ID
- Issue Title
- Constituency
- Issue Category
- Priority
- Status
- Representative

Reports

- Total Issues Report
- Pending Issues Report
- Resolved Issues Report
- High-Priority Issues Report
- Constituency-Wise Issue Report
- Category-Wise Issue Report
- Student-Wise Issue Report
- Representative Assignment Report
- Issue Response Report
- Status History Report

12. Application Interface / PPT Content

The proposed application flow can be presented as:

Application Flow

Login → Dashboard → Students → Constituencies → Submit Issue → Assign Issue → Track Status → Responses → Notifications → Reports

Main Modules

- Student Management
- Department Management
- Constituency Management
- MLA Management
- Issue Category Management
- Issue Management

- Representative Management
- Assignment Management
- Response Management
- Notification Management
- Reports & Analytics

13. Conclusion

CivicConnect demonstrates how a relational database can be used to organize and manage student civic and public grievance information.

The project contains 11 related tables connected through primary and foreign keys. It demonstrates important DBMS concepts including CRUD operations, joins, subqueries, normalization up to 3NF, views, triggers, stored procedures and reporting.

The system provides a structured approach for submitting, assigning, tracking and responding to civic issues while maintaining proper database relationships and historical records.